

26.1

Cyclomatic Complexity (4)

26.1.1 Cyclomatic Complexity: NIELIT 2021 Dec Scientist B - Section B: 39



A particular control flow graph has 15 nodes out of which 7 are decision nodes. The cyclomatic complexity and number of edges respectively can be given by the following :

- A. 7,21 B. 15,24 C. 8,21 D. 16,16

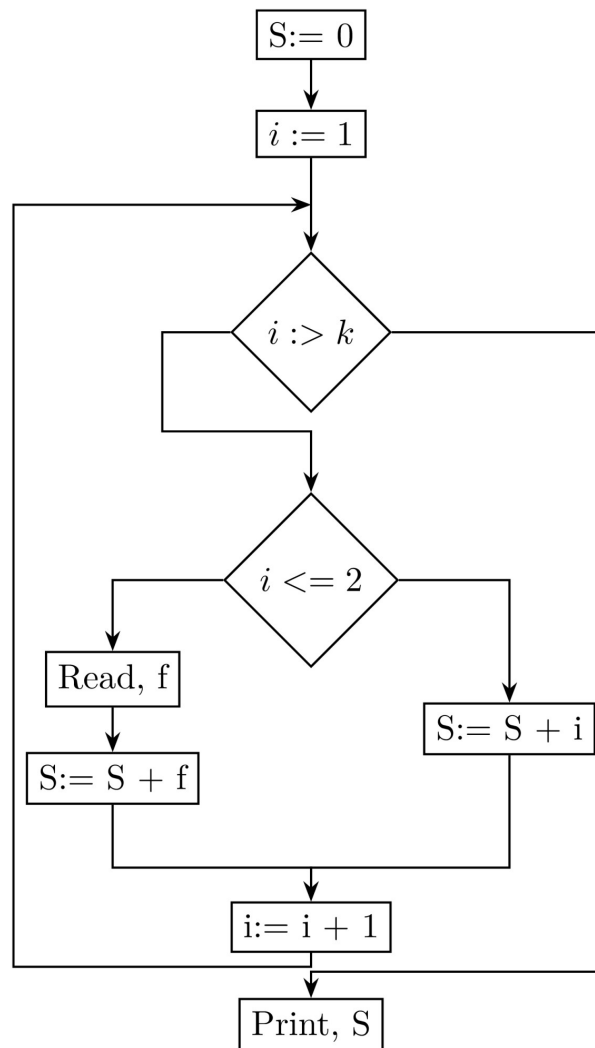
nielit-2021-it-dec-scientistb is&software-engineering cyclomatic-complexity

Answer key

26.1.2 Cyclomatic Complexity: NIELIT 2022 April Scientist B | Section B | Question: 84



To carry out white box testing of a program, its flow chart representation is obtained as shown in the figure below:



For basis path based testing of this program, its cyclomatic complexity is

A. 5

B. 4

C. 3

D. 2

nielit2022apr-scientistb is&software-engineering cyclomatic-complexity normal

Answer key **26.1.3 Cyclomatic Complexity: NIELIT Scientific Assistant A 2020 November: 113**

Consider the following C program segment.



```

while(first<=last)
{
    if(array[middle]<search)
        first=middle+1;
    else if(array[middle]== search)
        found=True;
    else last=middle-1;
    middle=(first+last)/2;
}
if(first<last)not Present = True;

```

The cyclomatic complexity of the program segment is _____.

A. 3

B. 4

C. 5

D. 6

nielit-sta-2020 is&software-engineering cyclomatic-complexity

Answer key **26.1.4 Cyclomatic Complexity: NIELIT Scientist B 2020 November: 53**

Which one of the following statements is incorrect?

- A. The number of regions corresponds to the cyclomatic complexity
- B. Cyclomatic complexity for a flow graph G is $V(G) = N - E + 2$, where E is the number of edges and N is the number of nodes in flow graph
- C. Cyclomatic complexity for a flow graph G is $V(G) = E - N + 2$, where E is the number of edges and N is the number of nodes in flow graph
- D. Cyclomatic complexity for a flow graph G is $V(G) = P + 1$, where P is the number of predicate nodes contained in the flow graph G

nielit-scb-2020 cyclomatic-complexity software-testing is&software-engineering

Answer key **26.2****Data Flow Diagram (1)****26.2.1 Data Flow Diagram: NIELIT 2021 Dec Scientist B - Section B: 53**

The _____ enables the software engineer to develop models of the information domain and functional domain at the same time.

A. data flow diagram

B. state transition diagram

C. control specification

D. activity diagram

Answer key

26.3

Function Point Metric (1)

26.3.1 Function Point Metric: NIELIT 2017 July Scientist B (IT) - Section B: 27



The Function Points(FP) calculated for software projects are often used to obtain an estimate of Lines of Code(LOC) required for that project. Which of the following statements is FALSE in this context?

- A. The relationship between FP and LOC depends on the programming language used to implement the software.
- B. LOC requirement for an assembly language implementation will be more for a given FP value, than LOC for implementation in COBOL.
- C. On an average, one LOC of C++ provides approximately 1.6 times the functionality of a single LOC of FORTRAN.
- D. FP and LOC are not related to each other.

Answer key

26.4

Functions (1)

26.4.1 Functions: NIELIT 2022 April Scientist B | Section B | Question: 98



Consider a software project with the following information domain characteristics for calculation of function point metric.

- Number of external inputs (I) = 30
- Number of external outputs (O) = 60
- Number of external inquiries (E) = 23
- Number of files (F) = 08
- Number of external interfaces (N) = 02\$

It is given that the complexity weighting factors for I, O, E, F and N are 4, 5, 4, 10 and 7, respectively. It is also given that, out of fourteen value adjustment factors that influence the development effort, four factors are not applicable, each of the other four factors have value 3, and each of the remaining factors have value 4. The computed value of function point metric is _____.

- A. 612.06 B. 212.05 C. 305.09 D. 806.9

26.5

Is&software Engineering (45)

26.5.1 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 18

Which of the following is not defined in a good Software Requirement Specification(SRS)

document?



- A. Functional requirement
- B. Goals of implementation
- C. Nonfunctional requirement
- D. Algorithm for software implementation

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.2 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 29

Software Requirement Specification (SRS) is also known as specification of



- A. White box testing
- B. Integrated testing
- C. Acceptance testing
- D. Black box testing

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.3 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 32

Line of code(LOC) of the product comes under which type of measures?



- A. Indirect measures
- B. Coding
- C. Direct measures
- D. None of the above

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.4 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 38

What is the testing to ensure the WebApp properly interfaces with other applications or databases?



- A. Compatibility
- B. Interoperability
- C. Performance
- D. Security

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.5 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 44

What is described by means of DFDs as studied earlier and represented in algebraic form?



- A. Data flow
- B. Data storage
- C. Data structures
- D. Data elements

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.6 Is&software Engineering: NIELIT 2016 DEC Scientist B (IT) - Section B: 32

Software projects management comprises of a number of activities, which contains:



- A. Project planning
- B. Project estimation
- C. Scope management
- D. All mentioned above

nielit2016dec-scientistb-it non-gatecse is&software-engineering

Answer key

26.5.7 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 46



According to Brooks, if n is the number of programmers in a project team, then the number of communication path is

- A. $n(n-1)/2$
- B. $n \log n$
- C. n
- D. $n(n+1)/2$

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key

26.5.8 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 47



In object oriented design of software, objects have

- A. attributes and name only
- B. operations and name only
- C. attributes name and operations
- D. mutation and permutation property

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key

26.5.9 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 49



The extent to which the software can control to operate correctly despite the introduction of invalid input is called as

- A. reliability
- B. robustness
- C. fault tolerance
- D. portability

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key

26.5.10 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 50



On an average, the programmer months is given by $3.6 \times (KLOC)^{1.2}$. If so, a project requiring one thousand source instructions will require

- A. 3.6 PM.
- B. 0.36 PM.
- C. 0.0036 PM.
- D. 7.23 PM

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key

26.5.11 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 51

Relation of COCOMO model is



A. $E = a * (KLOC)^6$ B. $E = a * (KLOC)^5$ C. $E = a * (KLOC)^4$ D. $E = a * (KLOC)^3$

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key

26.5.12 Is&software Engineering: NIELIT 2016 MAR Scientist C - Section C: 1



The advantage of better testing in software development is in

A. waterfall model B. prototyping C. iterative D. all of these

nielit2016mar-scientistc non-gatecse is&software-engineering

Answer key

26.5.13 Is&software Engineering: NIELIT 2016 MAR Scientist C - Section C: 23



The graph theoretic concept will be useful in software testing is

A. Cyclomatic number B. Hamiltonian circuit
C. Eulerian cycle D. None of these

nielit2016mar-scientistc non-gatecse is&software-engineering

Answer key

26.5.14 Is&software Engineering: NIELIT 2016 MAR Scientist C - Section C: 24



In testing phase, the effort distribution is upto

A. 10% B. 20% C. 40% D. 50%

nielit2016mar-scientistc non-gatecse is&software-engineering

Answer key

26.5.15 Is&software Engineering: NIELIT 2017 DEC Scientist B - Section B: 29



Which of the following is not deliverable of the structured system analysis?

A. Data flow diagram B. Prototype model
C. Entity Relationship diagram D. Data dictionary

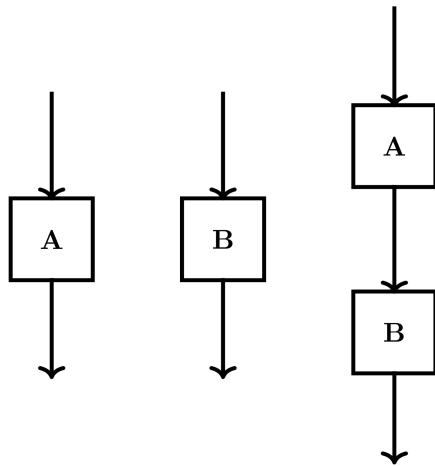
nielit2017dec-scientistb non-gatecse is&software-engineering

Answer key

26.5.16 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 44



The Cyclomatic complexity of each of the modules A and B shown below is 10. What is the Cyclomatic complexity of the sequential integration shown on the right hand side?



A. 19

B. 21

C. 20

D. 10

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.17 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 45

What is the appropriate pairing of items in the two columns listing various activities encountered in a software life cycle?



- | | |
|-------------------------|--|
| P. Requirements Capture | 1. Module Development and integration |
| Q. Design | 2. Domain Analysis |
| R. Implementation | 3. Structural and Behavioral Marketing |
| S. Maintenance | 4. Performance Tunit |

- A. $P - 3, Q - 2, R - 4, S - 1$
 B. $P - 2, Q - 3, R - 1, S - 4$
 C. $P - 3, Q - 2, R - 1, S - 4$
 D. $P - 2, Q - 3, R - 4, S - 1$

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.18 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 51

A company needs to develop a strategy for software product development for which it has a choice of two programming languages $L1$ and $L2$. The number of Lines Of Code (LOC) developed using $L2$ is estimated to be twice the LOC developed with $L1$. The product will have to be maintained for five years. Various parameters for the company are given in the table below:



Parameter	Language L1	Language L2
Man years needed for development	LOC/10000	LOC/10000
Development cost per man year	Rs. 10,00,000	Rs. 7,50,000
Maintenance time	5 years	5 years
Cost of maintenance per year	Rs. 1,00,000	Rs. 50,000

Total cost of the project includes cost of development and maintenance. What is the LOC for $L1$ for which the cost of the project using $L1$ is equal to the cost of the project using $L2$?

- A. 4000 B. 5000 C. 4333 D. 4667

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key 

26.5.19 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 52

A company needs to develop digital signal processing software for one of its newest inventions. The software is expected to have 40000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.8 for the software development on embedded systems, while the exponentiation factor is given as 1.20. What is the estimated effort in person-months?



- A. 234.25 B. 932.50 C. 287.80 D. 122.40

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key 

26.5.20 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 53

Which one of the following is NOT desired in a good Software Requirement Specifications (SRS) document?



- A. Functional Requirements B. Non-Functional Requirements
C. Goals of Implementation D. Algorithms for Software Implementation

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key 

26.5.21 Is&software Engineering: NIELIT 2017 July Scientist B (IT) - Section B: 24

The coupling between different modules of a software is categorized as follows:



- I. Content coupling
- II. Common coupling
- III. Control coupling
- IV. Stamp Coupling

V. Data Coupling

Coupling between modules can be ranked in the order of strongest (least desirable) to weakest (most desirable) as follows:

- A. I-II-III-IV-V B. V-IV-III-II-I C. I-III-V-II-IV D. IV-II-V-III-I

nielit2017july-scientistb-it non-gatecse is&software-engineering

Answer key 

26.5.22 Is&software Engineering: NIELIT 2017 July Scientist B (IT) - Section B: 25

Which of the following statements are TRUE?



- I. The context diagram should depict the system as a single bubble.
- II. External entities should be identified clearly at all levels of DFDs.
- III. Control information should not be represented in a DFD.
- IV. A data store can be connected either to another data store or to an external entity.

- A. II and IV B. II and III C. I and III D. I, II and III

nielit2017july-scientistb-it non-gatecse is&software-engineering

Answer key 

26.5.23 Is&software Engineering: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 15

By open domain CASE tools we mean



- A. tools available in open domain
- B. software packages which can be downloaded from the internet
- C. software packages to aid each phase of the systems analysis and design which can be downloaded free of cost from the internet
- D. source codes of CASE tools

nielit2017oct-assistanta-cs non-gatecse is&software-engineering

Answer key 

26.5.24 Is&software Engineering: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 3

By open domain CASE tools we mean



- A. tools available in open domain
- B. software packages which can be downloaded from the internet
- C. software packages to aid each phase of the systems analysis and design which can be downloaded free of cost from the internet
- D. source codes of CASE tools

Answer key

26.5.25 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 100

Given the two statements S1 and S2 for software engineering :

S1 : Statement coverage cannot guarantee execution of loops in program under test.

S2 : Use of independent path testing criterion guarantees execution of each loop in a program under test more than once.

Then which among the following is true ?

- A. S1 is True, S2 is True
- B. S1 is True, S2 is False
- C. S1 is False, S2 is True
- D. S1 is False, S2 is False

nielit2021dec-scientista is&software-engineering



26.5.26 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 119

What is the main objective of ISO 9001 ?

- A. Verification
- B. Validation
- C. S/W Testing
- D. H/W Testing

nielit2021dec-scientista is&software-engineering

Answer key



26.5.27 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 69

What is the main focus of Reverse Engineering (RE) ?

- A. Data base structure
- B. S/W file structure
- C. Memory
- D. CPU Utilization

nielit2021dec-scientista is&software-engineering



26.5.28 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 81

Elicitation of requirements is a _____ .

- A. SDLC Process
- B. Cyclic Process
- C. SRS Process
- D. Development Process

nielit2021dec-scientista is&software-engineering

Answer key



26.5.29 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 93

Which of the following step is not a part of the requirement engineering process?

- A. Feasibility Study
- B. Programming Language Requirement Specification
- C. Software Requirement Specification
- D. Requirement Gathering & Validation



Answer key 

26.5.30 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 97



Which one of them is a good software?

- A. High cohesion low coupling
- B. Low cohesion high coupling
- C. High cohesion high coupling
- D. Low cohesion low coupling

Answer key 

26.5.31 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 100



Delivery of the software product on time is considered as _____ .

- A. SDLC
- B. User Satisfaction
- C. Planning
- D. UI/UX design for software

26.5.32 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 14



Efficiency in a software product does not include:

- A. responsiveness
- B. processing time
- C. memory utilization
- D. licensing

Answer key 

26.5.33 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 20



Build & Fix Model is suitable for programming exercises of how many LOC (Line of Code)?

- A. 100–200
- B. 200–400
- C. 400–1000
- D. above 1000

Answer key 

26.5.34 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 61



If developer wants to transform model into source code is also known as _____ .

- A. Backward Testing / Engineering
- B. Forward Engineering
- C. Forward Testing
- D. Reverse Engineering

Answer key

26.5.35 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 64



Which of these does not account for software failure?

- A. Increasing Demand
- B. Low expectation
- C. Increasing Supply
- D. Less reliable and expensive

nielit-2021-it-dec-scientistb is&software-engineering

26.5.36 Is&software Engineering: NIELIT 2022 April Scientist B | Section B | Question: 85



Match the following:

- 1) Waterfall model a) Specifications can be developed incrementally
- 2) Evolutionary model b) Inflexible partitioning of the project into stages
- 3) Component-based c) Explicit recognition of risk software engineering
- 4) Spiral development d) Requirement compromises are inevitable

- A. (1) – (a), (2) – (c), (3) – (b), (4) – (d)
- B. (1) – (b), (2) – (a), (3) – (d), (4) – (c)
- C. (1) – (d), (2) – (b), (3) – (a), (4) – (c)
- D. (1) – (c), (2) – (a), (3) – (b), (4) – (d)

nielit2022apr-scientistb is&software-engineering normal

Answer key

26.5.37 Is&software Engineering: NIELIT 2022 April Scientist B | Section B | Question: 89



A company needs to develop digital signal processing software for one of its newest inventions. The software is expected to have 40000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.8 for the software development on embedded systems, while the exponentiation factor is given as 1.20. What is the estimated effort in person-months?

- A. 234.25
- B. 932.50
- C. 287.80
- D. 122.40

nielit2022apr-scientistb is&software-engineering is&software-engineering

26.5.38 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 148



Which can be considered as a difficulty in changing the Attitude of the employees?

- A. Escalation of Commitment
- B. Providing new information

C. Use of fear

D. Influence by friends and peers

nielit2023feb-scientistc is&software-engineering logical-reasoning

26.5.39 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 145



What are three situational criteria identified in the Fiedler model?

- A. Job requirements, position power, and leadership ability
- B. Charisma, influence, and leader-member relations
- C. Leader-member relations, task structure, and position power
- D. Task structure, leadership ability, and group conflict

nielit2023feb-scientistd is&software-engineering

26.5.40 Is&software Engineering: NIELIT Scientific Assistant A 2020 November: 47



Which of the following is not a part of the Test Implementation and Execution Phase?

- A. Creating test suites from the test cases
- B. Executing test cases either manually or by using test execution tools
- C. Comparing actual results
- D. Designing the tests

nielit-sta-2020 is&software-engineering

Answer key

26.5.41 Is&software Engineering: NIELIT Scientific Assistant A 2020 November: 56



SRD stands for _____.

- | | |
|-------------------------------------|---------------------------------------|
| A. Software Requirements Definition | B. Structured Requirements Definition |
| C. Software Requirements Diagram | D. Structured Requirements Diagram |

nielit-sta-2020 is&software-engineering

Answer key

26.5.42 Is&software Engineering: NIELIT Scientific Assistant A 2020 November: 80



A software Requirements Specification (SRS) document should avoid discussing which one of the following?

- | | |
|--------------------------|---|
| A. User interface issues | B. Non-functional requirements |
| C. Design solutions | D. Interfaces with third party software |

nielit-sta-2020 is&software-engineering

Answer key



Consider the basic *COCOMO* model where E is the effort applied in person-months, D is the development time in chronological months, $KLOC$ is the estimated number of delivered lines of code (in thousands) and a_b, b_b, c_b, d_b have their usual meanings. The basic *COCOMO* equations are of the form.

- A. $E = a_b(KLOC) \exp(b_b), D = c_b(E) \exp(d_b)$
- B. $D = a_b(KLOC) \exp(b_b), E = c_b(D) \exp(d_b)$
- C. $E = a_b \exp(b_b), D = c_b(KLOC) \exp(d_b)$
- D. $E = a_b \exp(d_b), D = c_b(KLOC) \exp(b_b)$

nielit-scb-2020 is&software-engineering

Answer key



Consider a software project with the following information domain characteristic for calculation of function point metric.

Number of external inputs (I) = 30

Number of external output(O) = 60

Number of external inquiries (E) = 23

Number of files (F) = 08

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It is given that the complexity weighting factors for I, O, E, F and N are 4, 5, 4, 10 and 7, respectively. It is also given that, out of fourteen value adjustment factors that influence the development effort, four factors are not applicable, each of the other four factors have value 3, and each of the remaining factors have value 4. The computed value of function point metric is _____

- A. 612.06
- B. 212.05
- C. 305.09
- D. 806.9

nielit-scb-2020 is&software-engineering

Answer key



Match the following:

(1) Waterfall model	(a) Specifications can be developed incrementally
(2) Evolutionary model	(b) Re-usability in development
(3) Component based software engineering	(c) Explicit recognition of risk
(4) Spiral development	(d) Inflexible partitioning of the project into stages

- A. 1-a, 2-b, 3-c, 4-d
 B. 1-d, 2-a, 3-b, 4-c
 C. 1-d, 2-b, 3-a, 4-c
 D. 1-c, 2-a, 3-b, 4-d

nielit-scb-2020 is&software-engineering match-the-following non-gatecse

Answer key 

26.6

Productivity (1)

26.6.1 Productivity: NIELIT Scientist B 2020 November: 93



58000 LOC gaming software is developed with effort of 3 person-year. What is the productivity of person-month?

- A. 1.9 KLOC B. 1.6 KLOC C. 4.8 KLOC D. 4.2 KLOC

nielit-scb-2020 is&software-engineering productivity

Answer key 

26.7

Software Design (7)

26.7.1 Software Design: NIELIT 2016 DEC Scientist B (IT) - Section B: 29



RAD software process model stands for:

- A. Rapid Application Development. B. Rapid Application Design.
 C. Relative Application Development. D. Recent Application Development.

nielit2016dec-scientistb-it non-gatecse is&software-engineering software-design

Answer key 

26.7.2 Software Design: NIELIT 2016 DEC Scientist B (IT) - Section B: 38



What are the characteristics of software?

- A. Software is developed or engineered; It is not manufactured in the classical sense.
 B. Software can be custom built or custom build.
 C. Software doesn't "wear out".
 D. All mentioned above.

nielit2016dec-scientistb-it non-gatecse is&software-engineering software-design

Answer key 

26.7.3 Software Design: NIELIT 2021 Dec Scientist B - Section B: 68



Which type of illustration lists the functionality of whole project ?

- A. DFD–0
- B. Class Diagram
- C. Use case Diagram
- D. State Diagram

nielit2021dec-scientistb is&software-engineering software-design

Answer key 

26.7.4 Software Design: NIELIT Scientific Assistant A 2020 November: 104



Software consists of _____

- A. Set of instructions + operating procedures
- B. Programs + documentation + operating procedures
- C. Programs + hardware manuals
- D. Set of programs

nielit-sta-2020 is&software-engineering software-design

Answer key 

26.7.5 Software Design: NIELIT Scientific Assistant A 2020 November: 106



Changes are made to the system to reduce the future system failure chances is called _____.

- A. Preventive Maintenance
- B. Adaptive Maintenance
- C. Corrective Maintenance
- D. Perfective Maintenance

nielit-sta-2020 is&software-engineering software-design

Answer key 

26.7.6 Software Design: NIELIT Scientific Assistant A 2020 November: 116



In the context of modular software design, which one of the following combinations is desirable?

- A. High cohesion and high coupling
- B. High cohesion and low coupling
- C. Low cohesion and high coupling
- D. Low cohesion and low coupling

nielit-sta-2020 is&software-engineering software-design

Answer key 

26.7.7 Software Design: NIELIT Scientific Assistant A 2020 November: 81



In the Model-View-Controller (MVC) architecture, the model defines the _____.

- A. Data-access layer
- B. Presentation layer
- C. Business-logic layer
- D. Interface layer

nielit-sta-2020 is&software-engineering software-design

Answer key 

26.8.1 Software Development Life Cycle Models: NIELIT 2022 April Scientist B | Section B | Question: 53



What is the appropriate pairing of items in the two columns listing various activities encountered in a software life cycle?

- | | |
|-------------------------|--|
| P. Requirements Capture | 1. Module Development and integration |
| Q. Design | 2. Domain Analysis |
| R. Implementation | 3. Structural and Behavioral Marketing |
| S. Maintenance | 4. Performance Tunit |

- A. (P) – (3), (Q) – (2), (R) – (4), (S) – (1)
 B. (P) – (2), (Q) – (3), (R) – (1), (S) – (4)
 C. (P) – (3), (Q) – (2), (R) – (1), (S) – (4)
 D. (P) – (2), (Q) – (3), (R) – (4), (S) – (1)

nielit2022apr-scientistb is&software-engineering software-development-life-cycle-models

26.9.1 Software Testing: NIELIT 2017 July Scientist B (IT) - Section B: 28



The availability of complex software is 90% its Mean Time Between Failure(MTBF) is 200 days. Because of the critical nature of the usage, the organization deploying the software further enhanced it to obtain an availability of 95%. In the process, the Mean Time To Repair(MTTR) increased by 5 days. What is the MTBF of the enhanced software?(choose the nearest option)

- A. 205 days. B. 300 days. C. 500 days. D. 700 days.

nielit2017july-scientistb-it non-gatecse is&software-engineering software-testing

Answer key

26.9.2 Software Testing: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 21

A program P calls two subprograms $P1$ and $P2$. $P1$ can fail 50% time and $P2$ can fail 40% times. The program P can fail



- A. 50% B. 10% C. 60% D. 70%

nielit2017oct-assistanta-it is&software-engineering software-testing

Answer key

26.9.3 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 119

Integration testing, Unit Testing & System Testing are _____.



- A. Fundamental logic of Testing
- C. Core Testing

- B. Level Testing
- D. Testing Suites

nielit2021dec-scientistb software-testing is&software-engineering

Answer key 

26.9.4 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 44



Boundary value analysis is based on which testing :

- A. White Box Testing
- C. White Box & Black Box Testing
- B. Black Box Testing
- D. None of the options

nielit-2021-it-dec-scientistb is&software-engineering software-testing

Answer key 

26.9.5 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 45



Match the following:

List-I

- W. Condition coverage
- X. Equivalence class partitioning
- Y. Volume testing
- Z. Alpha Testing

List-II

- 1. Black-box testing
- 2. System testing
- 3. White-box testing
- 4. Performance testing

- A. W-2, X-3, Y-1, Z-4
- C. W-3, X-1, Y-4, Z-2
- B. W-3, X-4, Y-2, Z-1
- D. W-3, X-1, Y-2, Z-4

nielit2021dec-scientistb is&software-engineering software-testing

Answer key 

26.9.6 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 8



Test levels are performed in which order?

- A. Unit, Integration, Acceptance, System.
- C. Unit, Integration, System, Acceptance.
- B. It depends on the nature of a project.
- D. Unit, System, Integration, Acceptance.

nielit-2021-it-dec-scientistb software-testing is&software-engineering

Answer key 

26.9.7 Software Testing: NIELIT 2022 April Scientist B | Section B | Question: 50

Match the following:



List-I

- (P) Condition coverage
- (Q) Equivalence class partitioning
- (R) Volume testing
- (S) Alpha testing

List-II

- (1) Black-box testing
- (2) System testing
- (3) White-box testing
- (4) Performance testing

Codes :

	P	Q	R	S
(I)	(2)	(3)	(1)	(4)
(II)	(3)	(4)	(2)	(1)
(III)	(3)	(1)	(4)	(2)
(IV)	(3)	(1)	(2)	(4)

- A. (I) B. (II) C. (III) D. (IV)

nielit2022apr-scientistb is&software-engineering software-testing normal non-gatecse

Answer key 

26.9.8 Software Testing: NIELIT Scientific Assistant A 2020 November: 77



Black Box Software Testing method focuses on the :

- A. Boundary condition of the software B. Control structure of the software
C. Testing of user Interface only D. Cyclomatic Complexity

nielit-sta-2020 is&software-engineering software-testing

Answer key 

26.9.9 Software Testing: NIELIT Scientist B 2020 November: 116



Consider a software program that is artificially seeded with 100 faults. While testing this program, 159 faults are detected, out of which 75 faults are from those artificially seeded faults. Assuming that both real and seeded faults are of same nature and have same distribution, the estimated number of undetected real faults is _____.

- A. 28 B. 175 C. 56 D. 84

nielit-scb-2020 probability software-testing is&software-engineering

Answer key 

Answer Keys

26.1.1	C	26.1.2	C	26.1.3	C	26.1.4	B	26.2.1	A
26.3.1	D	26.4.1	A	26.5.1	D	26.5.2	D	26.5.3	C

26.5.4	B	26.5.5	A	26.5.6	D	26.5.7	A	26.5.8	C
26.5.9	B	26.5.10	A	26.5.11	X	26.5.12	B	26.5.13	A
26.5.14	D	26.5.15	B	26.5.16	A	26.5.17	B	26.5.18	B
26.5.19	A	26.5.20	D	26.5.21	A	26.5.22	C	26.5.23	C
26.5.24	C	26.5.25	A	26.5.26	A	26.5.27	B	26.5.28	C
26.5.29	B	26.5.30	A	26.5.31	B	26.5.32	D	26.5.33	A
26.5.34	B	26.5.35	C	26.5.36	B	26.5.37	A	26.5.38	A
26.5.39	C	26.5.40	D	26.5.41	B	26.5.42	C	26.5.43	A
26.5.44	A	26.5.45	N/A	26.6.1	B	26.7.1	A	26.7.2	D
26.7.3	C	26.7.4	B	26.7.5	A	26.7.6	B	26.7.7	A
26.8.1	B	26.9.1	C	26.9.2	D	26.9.3	B	26.9.4	B
26.9.5	C	26.9.6	B	26.9.7	C	26.9.8	C	26.9.9	A